

QUESTIONS 31 / 48 completed

Factorial

FACTORIAL

Write a program to read in an integer and print the factorial of that number using a for loop.

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     long long n;
6     cin >> n;
7
8     if (n < 0) {
9         cout << "Invalid input";
10        return 0;
11    }
12
13    long long fact = 1;
14    for (long long i = 1; i <= n; i++) {
15        fact *= i;
16    }
17
18    cout << fact;
19    return 0;
20 }
```

INPUT

Your INPUT goes here!

OUTPUT

QUESTIONS 31 / 48 completed

Prime

PRIME

Write a program to read in an integer and determine if it is a prime number or not using a for loop.

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     long long n;
6     cin >> n;
7
8     if (n <= 1) {
9         cout << "Not Prime";
10        return 0;
11    }
12
13    bool prime = true;
14    for (long long i = 2; i * i <= n; i++) {
15        if (n % i == 0) {
16            prime = false;
17            break;
18        }
19    }
20
21    cout << (prime ? "Prime" : "Not Prime");
```

INPUT

Your INPUT goes here!

OUTPUT

QUESTIONS 31 / 48 completed

Fibonacci

FIBONACCI

Write a program to read in an integer and print the Fibonacci series up to that number using a for loop.

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     long long n;
6     cin >> n;
7
8     long long a = 0, b = 1;
9
10    for (; a <= n; ) {
11        cout << a;
12        if (a + b <= n) cout << " ";
13        long long next = a + b;
14        a = b;
15        b = next;
16    }
17    return 0;
18 }
```

INPUT

Your INPUT goes
here!

QUESTIONS 31 / 48 completed

Sum of All

SUM OF ALL

Write a program to read in an integer and print the sum of all its digits using a while loop.

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     long long n;
6     cin >> n;
7
8     if (n == 0) {
9         cout << 0;
10        return 0;
11    }
12
13    n = (n < 0) ? -n : n;
14    int sum = 0;
15
16    while (n > 0) {
17        sum += n % 10;
18        n /= 10;
19    }
20
21    cout << sum;
```

INPUT

Your INPUT goes
here!

QUESTIONS 31 / 48 completed

Palindrome

PALINDROME

Write a program to read in an integer and determine if it is a palindrome or not using a while loop.

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     long long n;
6     cin >> n;
7
8     long long original = n;
9     long long rev = 0;
10
11     while (n > 0) {
12         rev = rev * 10 + (n % 10);
13         n /= 10;
14     }
15
16     if (rev == original)
17         cout << "Palindrome";
18     else
19         cout << "Not Palindrome";
20
21     return 0;
```

INPUT

Your INPUT goes
here!